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### Mechanical seal device

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#### Abstract

A circulation pump mechanism of a mechanical seal device includes a rotor having mounting grooves formed therein, vanes provided in the mounting grooves, a plurality of biasing members provided in a space on a deep side with respect to the vane in each mounting groove and configured to bias each vane in a direction in which the vane is caused to project, a housing having an inner circumferential surface which is eccentric with respect to an outer circumferential surface of the rotor and on which a tip of each vane slides, a pump chamber demarcated on a leading side in a rotation direction of the rotor with respect to the tip of each vane between the outer circumferential surface of the rotor and the inner circumferential surface of the housing, and a communication flow path formed in the rotor and providing communication between the space and the pump chamber.

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| <b>Inventors:</b>            | <b>Nakatsuka; Masaya (Osaka, JP), Hayashi; Masashi (Osaka, JP), Fujinaga; Shigeyuki (Osaka, JP)</b> |
| <b>Applicant:</b>            | <b>PILLAR CORPORATION (Osaka, JP)</b>                                                               |
| <b>Family ID:</b>            | <b>1000008819695</b>                                                                                |
| <b>Assignee:</b>             | <b>PILLAR CORPORATION (Osaka, JP)</b>                                                               |
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## References Cited

### U.S. PATENT DOCUMENTS

| Patent No. | Issued Date | Patentee Name | U.S. Cl. | CPC          |
|------------|-------------|---------------|----------|--------------|
| 2628852    | 12/1952     | Voytech       | 165/47   | F16J 15/38   |
| 3495840    | 12/1969     | Wilk          | 277/408  | F16J 15/342  |
| 3746350    | 12/1972     | Mayer         | 277/408  | F16J 15/3404 |
| 3888495    | 12/1974     | Mayer         | 277/927  | F04D 29/043  |
| 4545588    | 12/1984     | Nagai         | 277/408  | F16J 15/346  |
| 4721311    | 12/1987     | Kakabaker     | 277/408  | F16J 15/162  |
| 5195867    | 12/1992     | Stirling      | 415/111  | F16J 15/3404 |
| 9347565    | 12/2015     | Berger        | N/A      | F16J 15/3404 |
| 11441571   | 12/2021     | Yoshikawa     | N/A      | F04D 29/128  |
| 11698080   | 12/2022     | Yoshikawa     | 415/111  | F04D 29/124  |

### FOREIGN PATENT DOCUMENTS

| Patent No.  | Application Date | Country | CPC |
|-------------|------------------|---------|-----|
| 105649983   | 12/2015          | CN      | N/A |
| 2001-059487 | 12/2000          | JP      | N/A |
| 2006-144682 | 12/2005          | JP      | N/A |
| 2022-018073 | 12/2021          | JP      | N/A |

### OTHER PUBLICATIONS

International Search Report issued in corresponding International Application No. PCT/JP2023/000288, dated Mar. 28, 2023 (5 pages). cited by applicant

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*Primary Examiner:* Foster; Nicholas L

*Attorney, Agent or Firm:* Osha Bergman Watanabe & Burton LLP

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## Background/Summary

## TECHNICAL FIELD

(1) The present invention relates to a mechanical seal device.

## BACKGROUND ART

(2) A mechanical seal device installed in a rotary machine includes a circulation pump mechanism that circulates a sealing liquid in order to enhance lubrication between sealing surfaces of a rotary sealing ring and a fixed sealing ring and to remove sliding heat generated between the sealing surfaces (see, for example, PATENT LITERATURE 1). A mechanical seal device of PATENT LITERATURE 1 includes a circulation pump mechanism that functions as a vane pump. The circulation pump mechanism includes a rotor that rotates integrally with a rotary shaft of a rotary machine, a plurality of vanes that freely project and retract from the outer circumferential surface of the rotor, springs that respectively bias the vanes in a direction in which the vanes are caused to project, and a plurality of pump chambers that are each demarcated between the adjacent vanes. Such a circulation pump mechanism can circulate a fixed volume of a sealing liquid each time the rotor makes one rotation, and thus can circulate the required flow volume of the sealing liquid even in a rotary machine that rotates at low speeds, such as an agitator.

## CITATION LIST

Patent Literature

(3) PATENT LITERATURE 1: Japanese Laid-Open Patent Publication No. 2022-18073

## SUMMARY OF THE INVENTION

### Technical Problem

(4) In the circulation pump mechanism of PATENT LITERATURE 1, the tip of each vane rotating together with the rotor slides on the inner circumferential surface of the housing which is eccentric with respect to the outer circumferential surface of the rotor. Accordingly, when the pressing force from the inner circumferential surface of the housing becomes stronger due to the rotation of the rotor, each vane retracts into the rotor against the biasing force of the spring. When the pressing force from the inner circumferential surface of the housing becomes weaker due to the rotation of the rotor, each vane having retracted into the rotor projects from the outer circumferential surface of the rotor by the biasing force of the spring. However, the pressure of the sealing liquid in the pump chamber also acts on the tip of each vane having retracted into the rotor. Therefore, if the pressure of the sealing liquid increases, there is a concern that each vane having retracted into the rotor does not project from the outer circumferential surface of the rotor.

(5) The present disclosure has been made in view of such circumstances, and an object of the present disclosure is to provide a mechanical seal device that can suppress interference with projection movement of a vane with respect to a rotor in a circulation pump mechanism.

### Solution to Problem

(6) (1) The present disclosure is directed to a mechanical seal device including: a seal case surrounding a rotary shaft of a rotary machine and separating a machine inner area and a machine outer area of the rotary machine from each other; at least one mechanical seal placed between the seal case and the rotary shaft; a sealing liquid chamber which is formed in the seal case and into which a sealing liquid for lubricating a sliding portion of the mechanical seal is introduced; and a circulation pump mechanism configured to circulate the sealing liquid in the sealing liquid chamber, wherein the circulation pump mechanism includes an annular rotor provided on the rotary shaft side so as to be integrally rotatable with the rotary shaft and having a mounting groove open at an outer circumferential surface thereof, a vane provided in the mounting groove and configured to freely project and retract with respect to the outer circumferential surface of the rotor, a biasing member provided in a space on a deep side with respect to the vane in the mounting groove and configured to bias the vane in a direction in which the vane is caused to project from the outer circumferential surface of the rotor, a housing having, on an inner circumference thereof, a sliding surface which is eccentric with respect to the outer circumferential surface of the rotor and on

which a tip of the vane slides as the rotor rotates, the housing being mounted to the seal case, a pump chamber which is demarcated on a leading side in a rotation direction of the rotor with respect to the tip of the vane between the outer circumferential surface of the rotor and the sliding surface of the housing and into and from which the sealing liquid is sucked and discharged, and a communication flow path formed in the rotor and providing communication between the space of the mounting groove and the pump chamber.

(7) In the mechanical seal device of the present disclosure, the space on the deep side with respect to the vane in the mounting groove of the rotor of the circulation pump mechanism communicates with the pump chamber demarcated on the leading side in the rotation direction of the rotor with respect to the tip of the vane, via the communication flow path. Therefore, the sealing liquid in the pump chamber flows into the space of the mounting groove via the communication flow path. Accordingly, the pressure of the sealing liquid in the space of the mounting groove and the pressure of the sealing liquid in the pump chamber cancel each other, so that substantially only the biasing force of the spring acts on the vane. Therefore, even if the pressure of the sealing liquid in the pump chamber acting on the tip of the vane increases, interference with the projection movement of the vane with respect to the rotor can be suppressed.

(8) (2) In the mechanical seal device of (1) above, preferably, the communication flow path has a plurality of flow path portions aligned in an axial direction of the rotor.

(9) In this case, the sealing liquid in the pump chamber can be quickly caused to flow into the space of the mounting groove via the plurality of flow path portions.

(10) (3) In the mechanical seal device of (1) or (2) above, preferably, the communication flow path is formed in the rotor independently of the mounting groove.

(11) In this case, when the sealing liquid flows in the communication flow path from the pump chamber toward the space of the mounting groove, the flow of the sealing liquid does not interfere with the projection movement of the vane. Accordingly, interference with the projection movement of the vane with respect to the rotor can be further inhibited.

(12) (4) In the mechanical seal device of (3) above, preferably, a cut groove having a V-shape in an axial view is formed at a location, facing the pump chamber, on the outer circumferential surface of the rotor, and the communication flow path is formed so as to be open at a side surface of the cut groove.

(13) In this case, the communication flow path can be formed more easily than in the case where the communication flow path is formed so as to be open at the outer circumferential surface of the rotor.

(14) (5) In the mechanical seal device of (3) or (4) above, preferably, the communication flow path has a flow path portion composed of a hole having a circular cross-section.

(15) In this case, the communication flow path can be formed even more easily.

#### Advantageous Effects of the Invention

(16) The mechanical seal device of the present disclosure can suppress interference with the projection movement of the vane with respect to the rotor in the circulation pump mechanism.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a cross-sectional view showing a mechanical seal device according to a first embodiment of the present disclosure.

(2) FIG. 2 is a cross-sectional view as seen in the direction of arrows I-I in FIG. 1.

(3) FIG. 3 is an enlarged cross-sectional view showing an area around 6 o'clock in FIG. 2 in a circulation pump mechanism.

(4) FIG. 4 is a perspective view showing a retaining member.

- (5) FIG. 5 is a perspective view showing a vane.
- (6) FIG. 6 is a side view of a first outer side plate of a housing as seen from a machine outer side.
- (7) FIG. 7 is a side view of a second outer side plate of the housing as seen from the machine outer side.
- (8) FIG. 8 is an enlarged perspective view showing an area around a cut groove of a rotor.
- (9) FIG. 9 is a cross-sectional view of a circulation pump mechanism of a mechanical seal device according to a second embodiment of the present disclosure.
- (10) FIG. 10 is an enlarged cross-sectional view showing an area around 6 o'clock in FIG. 9 in the circulation pump mechanism of the second embodiment.
- (11) FIG. 11 is an enlarged perspective view showing an area around a mounting groove of a rotor of the second embodiment.

## DETAILED DESCRIPTION

(12) Next, preferred embodiments of the present disclosure will be described with reference to the accompanying drawings. At least parts of the embodiments described below may be combined as desired.

### First Embodiment

(13) [Mechanical Seal Device]

(14) FIG. 1 is a cross-sectional view showing a mechanical seal device 1 according to a first embodiment of the present disclosure. The mechanical seal device 1 is provided in a rotary machine 50 such as a pump and an agitator that handles a sealed fluid. The mechanical seal device 1 includes a seal case 2, a first mechanical seal 6, a second mechanical seal 7, a sealing liquid chamber 8, and a circulation pump mechanism 30.

(15) The seal case 2 is fixed to a casing (not shown) of the rotary machine 50, and separate a machine inner area A and a machine outer area B of the rotary machine 50 from each other. The seal case 2 of the present embodiment has a first case body 3, a second case body 4, and a third case body 5 in order from the machine outer area B side toward the machine inner area A side. The first case body 3, the second case body 4, and the third case body 5 are each formed in an annular shape and surround a rotary shaft 51 of the rotary machine 50. Hereinafter, in the present specification, the machine inner area A side is referred to as machine inner side, and the machine outer area B side is referred to as machine outer side.

(16) The first mechanical seal 6 and the second mechanical seal 7 constitute a double-type mechanical seal and are placed between the seal case 2 and the rotary shaft 51 so as to be spaced at a predetermined interval in the axial direction of the rotary shaft 51 (hereinafter referred to simply as "axial direction"). The first mechanical seal 6 is placed on the machine inner side and seals between the machine inner area A and the sealing liquid chamber 8 (described later). The second mechanical seal 7 is placed on the machine outer side and seals between the machine outer area B and the sealing liquid chamber 8.

(17) The first mechanical seal 6 includes a spring retainer 11, springs 12, drive pins 13, a drive collar 14, a retainer 15, a rotary sealing ring 16, a stationary sealing ring 17, set screws 18, pins 19, an O-ring 20, and an O-ring 21.

(18) The spring retainer 11 is formed in an annular shape and is fixed via a sleeve 9 on the outer circumferential side of the rotary shaft 51. A plurality of screw holes 11a (only one is shown in FIG. 1) penetrating in the radial direction are formed in the spring retainer 11 so as to be spaced at predetermined intervals in the circumferential direction. The set screw 18 is screwed into each screw hole 11a, and the tip of the set screw 18 presses the outer circumferential surface of the sleeve 9. Accordingly, the spring retainer 11 is fixed to the rotary shaft 51 side by the set screws 18.

(19) A plurality of through holes 11b (only one is shown in FIG. 1) penetrating in the axial direction are formed in the spring retainer 11 so as to be spaced at predetermined intervals in the circumferential direction. The drive pin 13 is inserted into each through hole 11b and is held so as to be movable within a predetermined range in the axial direction relative to the spring retainer 11.

- (20) An end portion on the machine inner side of each drive pin **13** is fixed to the annular drive collar **14** which is mounted on the outer circumferential surface of the sleeve **9** on the machine inner side with respect to the spring retainer **11** so as to be movable in the axial direction. Specifically, a plurality of screw holes **14a** (only one is shown in FIG. **1**) penetrating in the axial direction are formed in the drive collar **14** so as to be spaced at predetermined intervals in the circumferential direction. An externally threaded portion **13a** formed on the end portion on the machine inner side of each drive pin **13** is screwed into each screw hole **14a**. Accordingly, the drive collar **14** is restricted from rotating relative to the spring retainer **11** while being held via the drive pin **13** so as to be movable in the axial direction relative to the spring retainer **11**.
- (21) The spring retainer **11** holds end portions of a plurality of the springs **12** (only one is shown in FIG. **1**) at predetermined intervals in the circumferential direction. Other end portions of the springs **12** are in contact with the end surface on the machine outer side of the drive collar **14**. Accordingly, the drive collar **14** is biased toward the machine inner side by the biasing forces of the springs **12**.
- (22) The annular retainer **15** is placed adjacent to the drive collar **14** on the machine inner side with respect to the drive collar **14**. The retainer **15** is mounted on the outer circumferential surface of the sleeve **9** so as to be movable in the axial direction. A plurality of engagement pins **15a** (only one is shown in FIG. **1**) projecting toward the machine outer side are provided on the end surface on the machine outer side of the retainer **15** so as to be spaced at predetermined intervals in the circumferential direction.
- (23) Each engagement pin **15a** is engaged with an engagement hole **14b** formed in the drive collar **14**. Accordingly, the retainer **15** is restricted from rotating relative to the drive collar **14** while being held together with the drive collar **14** so as to be movable in the axial direction relative to the rotary shaft **51**. The O-ring **20** made of rubber is provided on the inner circumference of the retainer **15** so as to seal (secondarily seal) between the outer circumferential surface of the sleeve **9** and the inner circumference of the retainer **15**.
- (24) The annular rotary sealing ring **16** is fitted and fixed on the machine inner side of the retainer **15**. The end surface on the machine inner side of the rotary sealing ring **16** is a sealing surface **16a**. The stationary sealing ring **17** is placed adjacent to the rotary sealing ring **16** on the machine inner side with respect to the rotary sealing ring **16**. The stationary sealing ring **17** is fitted and fixed to the inner circumferential side of the third case body **5**. The end surface on the machine outer side of the stationary sealing ring **17** is a sealing surface **17a** on which the sealing surface **16a** of the rotary sealing ring **16** slides.
- (25) A plurality of pin grooves **17b** (only one is shown in FIG. **1**) are formed on the machine inner side of the stationary sealing ring **17** so as to be spaced at predetermined intervals in the circumferential direction. The pin **19** which is fixed on the inner circumferential side of the third case body **5** so as to project toward the machine outer side is inserted into each pin groove **17b**. Accordingly, the stationary sealing ring **17** is restricted from rotating relative to the third case body **5**. The O-ring **21** made of rubber is provided on the outer circumference of the stationary sealing ring **17** so as to seal (secondarily seal) between the inner circumferential surface of the third case body **5** and the outer circumference of the stationary sealing ring **17**.
- (26) Similar to the first mechanical seal **6**, the second mechanical seal **7** includes a spring retainer **11**, springs **12**, drive pins **13**, a drive collar **14**, a retainer **15**, a rotary sealing ring **16**, a stationary sealing ring **17**, set screws **18**, pins **19**, an O-ring **20**, and an O-ring **21**. The spring retainer **11** to the O-ring **21** that are the components of the second mechanical seal **7** are arranged so as to be symmetrical with the spring retainer **11** to the O-ring **21** that are the components of the first mechanical seal **6** in the axial direction with the circulation pump mechanism **30** interposed therebetween.
- (27) Each pin **19** of the second mechanical seal **7** is fixed on the inner circumferential side of the first case body **3** so as to project toward the machine inner side. The O-ring **21** of the second

mechanical seal **7** seals (secondarily seals) between the stationary sealing ring **17** and the first case body **3**. The other configuration of the second mechanical seal **7** is the same as that of the first mechanical seal **6**, and thus the description thereof is omitted.

(28) The sealing liquid chamber **8** is an annular space formed in the seal case **2**. A sealing liquid for lubricating and cooling sliding portions between the sealing surfaces **16a** and **17a** of the first mechanical seal **6** and sliding portions between the sealing surfaces **16a** and **17a** of the second mechanical seal **7** is introduced into the sealing liquid chamber **8**. Water, an oil, a solvent, or the like, which cause no problem even when being mixed in the sealed fluid, are used as the sealing liquid, depending on the sealed fluid.

(29) In the second case body **4**, a supply path **4a** for supplying the sealing liquid from the outside to the sealing liquid chamber **8** is formed. In the first case body **3**, a discharge path **3a** for discharging the sealing liquid from the sealing liquid chamber **8** to the outside is formed. Accordingly, the sealing liquid supplied from the supply path **4a** to the sealing liquid chamber **8** during operation of the rotary machine **50** lubricates and cools each sliding portion of the first and second mechanical seals **6** and **7** and is discharged through the discharge path **3a** to the outside.

(30) [Circulation Pump Mechanism]

(31) FIG. **2** is a cross-sectional view as seen in the direction of arrows I-I in FIG. **1**. In FIG. **1** and FIG. **2**, the circulation pump mechanism **30** circulates the sealing liquid in the sealing liquid chamber **8** and is placed between the first mechanical seal **6** and the second mechanical seal **7** in the seal case **2**. The circulation pump mechanism **30** includes a rotor **31**, a plurality of vanes **32**, a plurality of springs (biasing members) **33**, a housing **34**, a plurality of pump chambers **38**, and retaining members **42**. In FIG. **2**, cross-sections of the rotary shaft **51**, the vanes **32**, and the retaining members **42** are omitted.

(32) The rotor **31** is composed of, for example, an annular member made of a metal such as stainless steel. The rotor **31** is fixed to the outer circumferential side of the sleeve **9**. Specifically, a key **39** fixed to the outer circumference of the sleeve **9** is engaged with a key groove **31a** formed on the inner circumferential side of the rotor **31**, thereby fixing the rotor **31** to the outer circumferential side of the sleeve **9**. Accordingly, the rotor **31** is mounted on the rotary shaft **51** side so as to be integrally rotatable with the rotary shaft **51**, and the rotary torque of the rotary shaft **51** is transmitted to the rotor **31** via the sleeve **9**.

(33) [Mounting Grooves]

(34) FIG. **3** is an enlarged cross-sectional view showing an area around 6 o'clock in FIG. **2** in the circulation pump mechanism **30**. In FIG. **2** and FIG. **3**, a plurality of mounting grooves **31c** (four in FIG. **2**) are formed on the rotor **31** so as to be open at an outer circumferential surface **31b** thereof and spaced at predetermined intervals in the circumferential direction. Each mounting groove **31c** is formed over the entirety in the axial direction of the rotor **31** (direction perpendicular to the drawing sheet of FIG. **3**).

(35) Each mounting groove **31c** has a first groove portion **31d** and a second groove portion **31e**. The first groove portion **31d** is formed on the bottom side (deep side) of the mounting groove **31c**. The second groove portion **31e** is formed on the opening side of the mounting groove **31c**. A groove width **W1** of the first groove portion **31d** is larger than a groove width **W2** of the second groove portion **31e**. The depth of the first groove portion **31d** is shallower than the depth of the second groove portion **31e**.

(36) [Retaining Members]

(37) The retaining member **42** is fitted into the first groove portion **31d** of each mounting groove **31c** from the axially outer side thereof. FIG. **4** is a perspective view showing the retaining member **42**. In FIG. **3** and FIG. **4**, each retaining member **42** is composed of a rectangular parallelepiped-shaped member made of a synthetic resin. The retaining member **42** of the present embodiment is made of, for example, a fluorine resin such as polytetrafluoroethylene (PTFE). The retaining member **42** is fitted to a pair of side surfaces **31d1** of the first groove portion **31d** with a tightening

allowance. Accordingly, the retaining member **42** is fixed to the first groove portion **31d**.

(38) The retaining member **42** has an insertion hole **42a** into which the spring **33** is inserted. A plurality of insertion holes **42a** (two in FIG. 4) are formed in the retaining member **42** so as to be aligned in the axial direction of the rotor **31**. Each insertion hole **42a** is formed so as to penetrate the retaining member **42** in the depth direction of the first groove portion **31d**. Each insertion hole **42a** communicates with the second groove portion **31e**.

(39) [Springs]

(40) A plurality of springs **33** are provided in each mounting groove **31c**. Specifically, a plurality of springs **33** (two in this case) are individually inserted into the plurality of insertion holes **42a** of the retaining member **42** fixed to each first groove portion **31d**. Each spring **33** is, for example, a conical coil spring made of a metal such as stainless steel. The conical coil spring **33** is inserted into the mounting groove **31c** such that the large diameter end thereof is located on the deep side. The large diameter end of the conical coil spring **33** penetrates the insertion hole **42a** and is in contact with the bottom surface of the first groove portion **31d**.

(41) The length in the longitudinal direction of the conical coil spring **33** in a most contracted state (state shown in FIG. 3) thereof is longer than the depth of the first groove portion **31d**. Therefore, the conical coil spring **33** has an exposed portion **33a** which is always exposed from the insertion hole **42a** of the retaining member **42** into the second groove portion **31e**. In the present embodiment, the small diameter end side of the conical coil spring **33** is the exposed portion **33a**.

(42) [Vanes]

(43) In FIG. 2 and FIG. 3, the vane **32** is inserted into the second groove portion **31e** of each mounting groove **31c**. Each vane **32** is composed of, for example, a flat plate-shaped member made of carbon. The length in the axial direction (direction perpendicular to the drawing sheet of FIG. 3) of the vane **32** is substantially the same as the length in the axial direction of the mounting groove **31c**. The vane **32** is slidable relative to a pair of side surfaces **31e1** of the second groove portion **31e**. Accordingly, the vane **32** freely projects and retracts with respect to the outer circumferential surface **31b** of the rotor **31** in the second groove portion **31e**.

(44) FIG. 5 is a perspective view showing the vane **32**. In FIG. 3 and FIG. 5, a recess **32c** is formed on a back surface **32b** (surface located on the deep side of the second groove portion **31e**) of the vane **32**. A plurality of recesses **32c** (two in FIG. 5) are formed on the vane **32** so as to be aligned in the axial direction of the rotor **31**.

(45) Each recess **32c** is formed with a size that allows a portion on the tip (small diameter end) side of the exposed portion **33a** to be inserted therein. Accordingly, the tip end sides of the exposed portions **33a**, of the plurality of conical coil springs **33**, which are exposed to the vane **32** side from the retaining member **42**, are individually inserted into the plurality of recesses **32c**.

(46) The tip of the exposed portion **33a** inserted into each recess **32c** is in contact with the bottom surface of the recess **32c**. Accordingly, the plurality of conical coil springs **33** in each mounting groove **31c** bias the vane **32** in a direction in which the vane **32** is caused to project from the outer circumferential surface **31b** of the rotor **31**. The other portion of the exposed portion **33a** of each conical coil spring **33** is exposed in a space **S1** of the second groove portion **31e**.

(47) The space **S1** is a space formed on the deep side of the vane **32** in the second groove portion **31e**. The volume of the space **S1** increases and decreases as the vane **32** projects and retracts with respect to the rotor **31**. Specifically, the volume of the space **S1** increases when the vane **32** projects from the rotor **31**, and the volume of the space **S1** decreases when the vane **32** retracts into the rotor **31**.

(48) [Housing]

(49) In FIG. 1 and FIG. 2, the housing **34** of the circulation pump mechanism **30** is mounted on the inner circumferential side of the seal case **2**. Specifically, the housing **34** is placed between a step surface **3b** of the first case body **3** and a projection end surface **4c** of the second case body **4**. The housing **34** has a cam ring **35**, and a first outer side plate **36** and a second outer side plate **37** placed

on both sides in the axial direction of the cam ring 35. The cam ring 35, the first outer side plate 36, and the second outer side plate 37 are restricted from rotating relative to the seal case 2 by a plurality of pins 40 (three in FIG. 2) which penetrate the cam ring 35, the first outer side plate 36, and the second outer side plate 37 in the axial direction.

(50) A plurality of springs 41 (only one in FIG. 1) are provided between the first outer side plate 36 and the second case body 4 so as to be spaced at predetermined intervals in the circumferential direction. Each spring 41 presses the first outer side plate 36 against the second case body 4 toward the machine outer side. Accordingly, the housing 34 is held between the step surface 3b of the first case body 3 and the projection end surface 4c of the second case body 4 in a state of being biased toward the machine outer side, and is movable toward the machine inner side against the biasing forces of the springs 41. Each spring 41 may be placed between the first case body 3 and the second outer side plate 37 and press the second outer side plate 37 against the first case body 3 toward the machine inner side.

(51) The cam ring 35 is composed of, for example, an annular member made of a metal such as stainless steel. The cam ring 35 is placed between the first case body 3 and the rotor 31. The length in the axial direction of the cam ring 35 is substantially the same as the length in the axial direction of the rotor 31. An outer circumferential surface 35a of the cam ring 35 is a circumferential surface having a center at an axis C1 of the rotary shaft 51.

(52) An inner circumferential surface 35b of the cam ring 35 is a circumferential surface having a center C2 at a position eccentric with respect to the axis C1 of the rotary shaft 51. In the present embodiment, the center C2 of the inner circumferential surface 35b of the cam ring 35 is eccentric on the upper side of FIG. 2 with respect to the axis C1 of the rotary shaft 51. Accordingly, the inner circumferential surface 35b of the cam ring 35 is eccentric with respect to the outer circumferential surface 31b of the rotor 31. Specifically, the inner circumferential surface 35b of the cam ring 35 is eccentric so as to be gradually separated from the outer circumferential surface 31b of the rotor 31 in a first angle range  $\theta 1$  from 6 o'clock to 12 o'clock in FIG. 2. In addition, the inner circumferential surface 35b of the cam ring 35 is eccentric so as to be gradually closer to the outer circumferential surface 31b of the rotor 31 in a second angle range  $\theta 2$  from 12 o'clock to 6 o'clock in FIG. 2.

(53) The inner circumferential surface 35b of the cam ring 35 is a sliding surface on which a tip 32a of each vane 32 slides as the rotor 31 rotates. Each time the rotor 31 makes one rotation in the direction of an arrow E, each vane 32 alternately repeats projection movement of projecting from the mounting groove 31c and retraction movement of retracting into the mounting groove 31c while sliding on the inner circumferential surface 35b of the cam ring 35. Each vane 32 of the present embodiment moves so as to project in the first angle range  $\theta 1$  and moves so as to retract in the second angle range  $\theta 2$ .

(54) In the first angle range  $\theta 1$ , each vane 32 gradually moves so as to project by the biasing forces of the springs 33 from a state of retracting most into the mounting groove 31c (6 o'clock state) to a state of projecting most from the mounting groove 31c (12 o'clock state). In the second angle range  $\theta 2$ , each vane 32 gradually moves so as to retract against the biasing forces of the springs 33 by being pressed against the inner circumferential surface 35b of the cam ring 35 from the state of projecting most (12 o'clock state) to the state of retracting most (6 o'clock state).

(55) The first outer side plate 36 and the second outer side plate 37 are each composed of, for example, a disc member made of carbon, and are placed between the first case body 3 and the sleeve 9. Both side surfaces in the axial direction of the rotor 31 slide on the side surface on the machine outer side of the first outer side plate 36 and the side surface on the machine inner side of the second outer side plate 37, respectively.

(56) Due to the above configuration, the plurality of pump chambers 38 (four in FIG. 2) are formed between the outer circumferential surface 31b of the rotor 31 and the inner circumferential surface 35b of the cam ring 35 in the interior of the housing 34 so as to be demarcated by the vanes 32, 32

adjacent to each other in the circumferential direction. Each pump chamber **38** is a space into and from which the sealing liquid is sucked and discharged.

(57) The volume of each pump chamber **38** changes as the rotor **31** rotates. In the present embodiment, each time the rotor **31** makes one rotation in the direction of the arrow E, the volume of the pump chamber **38** changes so as to gradually increase in the first angle range  $\theta 1$ , and the volume of the pump chamber **38** changes so as to gradually decrease in the second angle range  $\theta 2$ .

(58) FIG. **6** is a side view of the first outer side plate **36** of the housing **34** as seen from the machine outer side. As shown in FIG. **1** and FIG. **6**, a plurality of suction ports **36a** (three in FIG. **6**) are formed in the first angle range  $\theta 1$  of the first outer side plate **36** so as to be spaced at predetermined intervals in the circumferential direction. Accordingly, when each pump chamber **38** passes through the first angle range  $\theta 1$  due to the rotation of the rotor **31**, the volume of the pump chamber **38** gradually increases, thereby causing the pressure in the pump chamber **38** to change from a positive pressure to a negative pressure. When the pressure in the pump chamber **38** becomes negative as described above, an action of sucking the sealing liquid through each suction port **36a** into the pump chamber **38** is generated. Accordingly, the sealing liquid in the sealing liquid chamber **8** is sucked through each suction port **36a** into the pump chamber **38**.

(59) FIG. **7** is a side view of the second outer side plate **37** of the housing **34** as seen from the machine outer side. As shown in FIG. **1** and FIG. **7**, a plurality of discharge ports **37a** (three in FIG. **7**) are formed in the second angle range  $\theta 2$  of the second outer side plate **37** so as to be spaced at predetermined intervals in the circumferential direction. Accordingly, when each pump chamber **38** passes through the second angle range  $\theta 2$  due to the rotation of the rotor **31**, the volume of the pump chamber **38** gradually decreases, thereby causing the pressure in the pump chamber **38** to change from a negative pressure to a positive pressure. When the pressure in the pump chamber **38** becomes positive as described above, an action of discharging the sealing liquid through each discharge port **37a** to the outside of the pump chamber **38** is generated. Accordingly, the sealing liquid in the pump chamber **38** is discharged through each discharge port **37a** to the sealing liquid chamber **8**.

(60) Due to the above configuration, the circulation pump mechanism **30** functions as a vane pump, and can circulate a fixed volume of the sealing liquid each time the rotor **31** makes one rotation. In FIG. **1**, for ease of understanding, the positions of the suction port **36a** and the discharge port **37a** are shown so as to be shifted in the circumferential direction.

(61) [Cut Grooves]

(62) In FIG. **2** and FIG. **3**, a plurality of cut grooves **31f** are formed on the outer circumferential surface **31b** of the rotor **31** so as to be spaced in the circumferential direction. The number of cut grooves **31f** is the same as the number of vanes **32** (four). Each cut groove **31f** is formed on the outer circumferential surface **31b** of the rotor **31** at a position, on the leading side in the rotation direction of the rotor **31**, slightly further away from the tip **32a** of the vane **32**. Each cut groove **31f** is formed in a V-shape in an axial view.

(63) Each cut groove **31f** has a first side surface **31g** and a second side surface **31h**. The first side surface **31g** is formed on the trailing side in the rotation direction of the rotor **31**. The second side surface **31h** is formed in the leading side in the rotation direction of the rotor **31**. An angle  $\alpha$  formed between the first side surface **31g** and the second side surface **31h** is set to have a magnitude that allows a machining tool (such as a drill) to easily access the first side surface **31g** when a communication flow path **45** described later is machined in the rotor **31** with the machining tool.

(64) [Communication Flow Paths]

(65) The circulation pump mechanism **30** further includes a plurality of communication flow paths **45** formed in the rotor **31**. The number of communication flow paths **45** is the same as the number of vanes **32** (four). Each communication flow path **45** is a flow path that provides communication between the space **S1** of the mounting groove **31c** and the pump chamber **38**. Each communication flow path **45** of the present embodiment is formed in the rotor **31** independently of the mounting

groove **31c**.

(66) FIG. **8** is an enlarged perspective view showing an area around the cut groove **31f** of the rotor **31**. In FIG. **3** and FIG. **8**, each communication flow path **45** has a plurality of flow path portions **46** aligned so as to be spaced in the axial direction of the rotor **31** (up-down direction in FIG. **8**). Each flow path portion **46** is composed of, for example, a hole having a circular cross-section. One end in the longitudinal direction of each flow path portion **46** is open at the first side surface (side surface) **31g** of the cut groove **31f**. The other end in the longitudinal direction of each flow path portion **46** is open at the side surface **31e1** of the second groove portion **31e** of the mounting groove **31c**. Due to the above, each flow path portion **46** of each communication flow path **45** provides communication between the space **S1** of the mounting groove **31c** and the pump chamber **38** demarcated on the leading side in the rotation direction of the rotor **31** with respect to the tip **32a** of the vane **32** inserted into the mounting groove **31c**.

(67) In FIG. **2** and FIG. **3**, when the rotor **31** rotates in the first angle range  $\theta 1$ , the plurality of flow path portions **46** function as supply flow paths for supplying the sealing liquid from the pump chamber **38** into the space **S1** of the mounting groove **31c**. Specifically, in the first angle range  $\theta 1$ , the sealing liquid is sucked through each suction port **36a** of the first outer side plate **36** into the pump chamber **38**. Accordingly, the sealing liquid in the pump chamber **38** flows from the cut groove **31f**, which is formed on the outer circumferential surface **31b** of the rotor **31** facing the pump chamber **38**, into the plurality of flow path portions **46** which are open at the first side surface **31g** of the cut groove **31f**. The sealing liquid having flowed into the plurality of flow path portions **46** flows into the space **S1** of the mounting groove **31c** communicating with these flow path portions **46**. Therefore, when the rotor **31** rotates in the first angle range  $\theta 1$ , the pressure of the sealing liquid having flowed into the space **S1** acts as a back pressure on the back surface **32b** of the vane **32**.

(68) When the rotor **31** rotates in the second angle range  $\theta 2$ , the plurality of flow path portions **46** function as discharge flow paths for discharging the sealing liquid in the space **S1** to the pump chamber **38**. Specifically, in the second angle range  $\theta 2$ , the vane **32** moves so as to retract into the mounting groove **31c** against the biasing forces of the conical coil springs **33** as described above. Accordingly, the sealing liquid in the space **S1** passes through the plurality of flow path portions **46** and is discharged from the cut groove **31f** to the pump chamber **38**. At that time, dust in the space **S1** is discharged together with the sealing liquid to the pump chamber **38** via the plurality of flow path portions **46**, so that the plurality of flow path portions **46** also function as flow paths for cleaning the space **S1**.

#### Advantageous Effects

(69) As described above, in the mechanical seal device **1** of the present embodiment, the space **S1** on the deep side with respect to the vane **32** in each mounting groove **31c** of the rotor **31** of the circulation pump mechanism **30** communicates with the pump chamber **38** demarcated on the leading side in the rotation direction of the rotor **31** with respect to the tip **32a** of the vane **32**, via the communication flow path **45**. Therefore, the sealing liquid in the pump chamber **38** flows into the space **S1** of the mounting groove **31c** via the communication flow path **45**. Accordingly, the pressure of the sealing liquid in the space **S1** of the mounting groove **31c** and the pressure of the sealing liquid in the pump chamber **38** cancel each other, so that substantially only the biasing forces of the springs **33** act on the vane **32**. Therefore, even if the pressure of the sealing liquid in the pump chamber **38** acting on the tip **32a** of the vane **32** increases, interference with the projection movement of the vane **32** with respect to the rotor **31** can be suppressed.

(70) Since each communication flow path **45** has the plurality of flow path portions **46** aligned in the axial direction of the rotor **31**, the sealing liquid in the pump chamber **38** can be quickly caused to flow into the space **S1** of the mounting groove **31c** via the plurality of flow path portions **46**.

(71) Each communication flow path **45** is formed in the rotor **31** independently of the mounting groove **31c**. Therefore, when the sealing liquid flows in the communication flow path **45** from the

pump chamber **38** toward the space **S1** of the mounting groove **31c**, the flow of the sealing liquid does not interfere with the projection movement of the vane **32**. Accordingly, interference with the projection movement of the vane **32** with respect to the rotor **31** can be further inhibited. In addition, the longitudinal direction of each communication flow path **45** is closer to a tangential direction of the outer circumferential surface **31b** of the rotor **31** than the longitudinal direction of each communication flow path **45** of a second embodiment described later, so that the sealing liquid in the pump chamber **38** can be efficiently caused to flow into the space **S1** of the mounting groove **31c** via the communication flow path **45**.

(72) Each communication flow path **45** is formed so as to be open at the first side surface **31g** of the V-shaped cut groove **31f** which is formed on the outer circumferential surface **31b** of the rotor **31**. Therefore, the communication flow path **45** can be formed more easily than in the case where the communication flow path **45** is formed so as to be open at the outer circumferential surface **31b** of the rotor **31**.

(73) Since each flow path portion **46** of each communication flow path **45** is composed of a hole having a circular cross-section, the communication flow path **45** can be formed even more easily.

Second Embodiment

(74) FIG. **9** is a cross-sectional view showing a circulation pump mechanism **30** of a mechanical seal device **1** according to the second embodiment of the present disclosure. FIG. **10** is an enlarged cross-sectional view showing an area around 6 o'clock in FIG. **9** in the circulation pump mechanism **30**. The circulation pump mechanism **30** of the present embodiment is different from the first embodiment in that the retaining members **42** (see FIG. **2**) into which the springs **33** are inserted are not included, and each of the configurations of the mounting grooves **31c** of the rotor **31** and the communication flow paths **45** is different.

(75) In FIG. **9** and FIG. **10**, each mounting groove **31c** of the rotor **31** in the circulation pump mechanism **30** of the present embodiment has a constant groove width **W3** over the entirety in the depth direction thereof (see also FIG. **11**). Each mounting groove **31c** has a first side surface **31i** and a second side surface **31j** facing each other. The first side surface **31i** is formed on the leading side in the rotation direction of the rotor **31**. The second side surface **31j** is formed on the trailing side in the rotation direction of the rotor **31**.

(76) FIG. **11** is an enlarged perspective view showing an area around the mounting groove **31c** of the rotor **31**. In FIG. **10** and FIG. **11**, a plurality of small grooves **31k** (three in FIG. **11**) are formed on the first side surface **31i** of each mounting groove **31c** so as to be aligned in the axial direction of the rotor **31**. Each small groove **31k** is formed to have an arc-shaped cross-section. Each small groove **31k** is formed over the entirety in the depth direction of the mounting groove **31c**. Each small groove **31k** is formed with a size that allows a portion in the circumferential direction at the large diameter end of the conical coil spring **33** to be inserted thereinto.

(77) The conical coil spring **33** is inserted into each mounting groove **31c** such that a portion in the circumferential direction of the conical coil spring **33** is placed in each of the plurality of small grooves **31k**. The vane **32** is inserted at the front of the conical coil spring **33** in each mounting groove **31c**. The vane **32** is slidable relative to the first side surface **31i** and the second side surface **31j** of the mounting groove **31c**. The small diameter end of the conical coil spring **33** is in contact with the back surface **32b** of the vane **32**.

(78) Due to the above configuration, a plurality of conical coil springs **33** (three in this case) are provided in a space **S2** on the deep side with respect to the vane **32** in each mounting groove **31c** so as to be aligned in the axial direction. These conical coil springs **33** bias the vane **32** in a direction in which the vane **32** is caused to project from the outer circumferential surface **31b** of the rotor **31**. Each conical coil spring **33** of the present embodiment is inserted into the mounting groove **31c** such that the large diameter end thereof is placed on the deep side, but may be inserted into the mounting groove **31c** such that the small diameter end thereof is placed on the deep side.

(79) In FIG. **9** and FIG. **10**, the plurality of small grooves **31k** formed in each mounting groove **31c**

communicate with the space S2 on the deep side of the mounting groove 31c. In addition, the plurality of small grooves 31k are open at the outer circumferential surface 31b of the rotor 31. Accordingly, the plurality of small grooves 31k provide communication between the space S2 of each mounting groove 31c and the pump chamber 38 demarcated on the leading side in the rotation direction of the rotor 31 with respect to the mounting groove 31c. Therefore, in the present embodiment, the plurality of small grooves 31k formed in each mounting groove 31c function as the plurality of flow path portions 46 of the communication flow path 45.

(80) Each small groove 31k has both a function of placing a portion in the circumferential direction of the conical coil spring 33 therein and a function as the flow path portion 46, but may function only as the flow path portion 46. In this case, each conical coil spring 33 may be sized to be inserted within the groove width W3 of the mounting groove 31c. The other components of the present embodiment are the same as in the first embodiment, and thus are designated by the same reference signs, and the description thereof is omitted.

(81) As described above, in the mechanical seal device 1 of the present embodiment as well, the sealing liquid in each pump chamber 38 flows into the space S2 of the mounting groove 31c via the communication flow path 45. Accordingly, the pressure of the sealing liquid in the space S2 of the mounting groove 31c and the pressure of the sealing liquid in the pump chamber 38 cancel each other, so that substantially only the biasing forces of the springs 33 act on the vane 32. Therefore, even if the pressure of the sealing liquid in the pump chamber 38 acting on the tip 32a of the vane 32 increases, interference with the projection movement of the vane 32 with respect to the rotor 31 can be suppressed.

(82) Since each communication flow path 45 has the plurality of flow path portions 46 (small grooves 31k) aligned in the axial direction of the rotor 31, the sealing liquid in the pump chamber 38 can be quickly caused to flow into the space S2 of the mounting groove 31c via the plurality of flow path portions 46.

Others

(83) The mechanical seal device 1 of the present disclosure includes a double-type mechanical seal, but it is sufficient that the mechanical seal device 1 includes at least one mechanical seal such as a single-type mechanical seal. The circulation pump mechanism 30 does not have to include the retaining members 42 into which the biasing members 33 are inserted.

(84) The number of biasing members 33 provided in each mounting groove 31c and the number of flow path portions 46 of each communication flow path 45 are not limited to those of the above embodiments. Each biasing member 33 is not limited to a conical coil spring, and may be, for example, a cylindrical spring.

(85) The embodiments disclosed herein are merely illustrative and not restrictive in all aspects. The scope of the present invention is defined by the scope of the claims rather than the meaning described above, and is intended to include meaning equivalent to the scope of the claims and all modifications within the scope.

#### REFERENCE SIGNS LIST

(86) 1 mechanical seal device 2 seal case 6 first mechanical seal (mechanical seal) 7 second mechanical seal (mechanical seal) 8 sealing liquid chamber 30 circulation pump mechanism 31 rotor 31b outer circumferential surface 31c mounting groove 31f cut groove 31g first side surface (side surface) 32 vane 32a tip 33 spring (biasing member) 34 housing 35b inner circumferential surface (sliding surface) 38 pump chamber 45 communication flow path 46 flow path portion 50 rotary machine 51 rotary shaft A machine inner area B machine outer area S1, S2 space

## Claims

1. A mechanical seal device comprising: a seal case surrounding a rotary shaft of a rotary machine and separating a machine inner area and a machine outer area of the rotary machine from each

other; at least one mechanical seal placed between the seal case and the rotary shaft; a sealing liquid chamber which is formed in the seal case and into which a sealing liquid for lubricating a sliding portion of the mechanical seal is introduced; and a circulation pump mechanism configured to circulate the sealing liquid in the sealing liquid chamber, wherein the circulation pump mechanism includes an annular rotor provided on the rotary shaft side so as to be integrally rotatable with the rotary shaft and having a mounting groove open at an outer circumferential surface thereof, a vane provided in the mounting groove and configured to freely project and retract with respect to the outer circumferential surface of the rotor, a biasing member provided in a space on a deep side with respect to the vane in the mounting groove and configured to bias the vane in a direction in which the vane is caused to project from the outer circumferential surface of the rotor, a housing having, on an inner circumference thereof, a sliding surface which is eccentric with respect to the outer circumferential surface of the rotor and on which a tip of the vane slides as the rotor rotates, the housing being mounted to the seal case, a pump chamber which is demarcated on a leading side in a rotation direction of the rotor with respect to the tip of the vane between the outer circumferential surface of the rotor and the sliding surface of the housing and into and from which the sealing liquid is sucked and discharged, and a communication flow path formed in the rotor and providing communication between the space of the mounting groove and the pump chamber.

2. The mechanical seal device according to claim 1, wherein the communication flow path has a plurality of flow path portions aligned in an axial direction of the rotor.

3. The mechanical seal device according to claim 1, wherein the communication flow path is formed in the rotor independently of the mounting groove.

4. The mechanical seal device according to claim 3, wherein a cut groove having a V-shape in an axial view is formed at a location, facing the pump chamber, on the outer circumferential surface of the rotor, and the communication flow path is formed so as to be open at a side surface of the cut groove.

5. The mechanical seal device according to claim 3, wherein the communication flow path has a flow path portion composed of a hole having a circular cross-section.

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